

ENVIRONMENTAL MODIFICATION STANDARDS

GENERAL RAMP SPECIFICATIONS

These specifications are to be used as a guide for HCBS providers. They must not be used to diminish in any way local, state and national codes. These specifications are a summary of information gathered using the International Building Code (2010 Edition), ICC, ANSI A117.1 1998 and ADA Standards for Accessible Design.

A. General Notes

- Any applicable building permits are the responsibility of the provider
- Ramps are to be firm, stable, slip resistant and safe.
- Ramps and landings with drop offs must have a minimum of 4" curb and railings.
- Ramps must be built using pressure (weather) treated lumber.
- Please advise participants that ramps need to be treated/weatherproofed annually.
- Ramps and landings are measured by the clear space provided, 56"x56" landings must not be counted as 5'x5' landings and must be re-built.
- No toenailing of railings between posts is permitted.
- All screws must be predrilled with pilot holes to prevent splitting.
- There is only one type/size of ramp, no heavier duty.

B. General Materials

- 4x4 posts
- 2x6 (min.) joists
- 2x6 or 5/4x6 decking
- 2x4 (min.) curbing, top, and top support rail
- 2x2 pickets
- 1 ¼" galvanized pipes (fence top rails) are to be used as handrails with galvanized "C" clamps and PVC caps.
- 2x4x4" or 2x4x6 handrail spacers
- 3" ring-shanked exterior nails and 3" and 2½" decking screws
- 2500 PSI cement

C. Foundations

- Posts must be a minimum 4x4 CCA lumber.
- Posts must be sunk a minimum 12" into the ground and set in concrete (min. 25lbs).
- Exception – if post is set on cement walkway or driveway, post may be anchored with a post base attached to the cement and/or cross- braced for stability.
- Foundation posts are to be set no more than 8' apart.
- Set the final post back 8" from the end of the ramp.

D. Floor Framing

- Outside joists shall consist of 2x6 (min) lumber connected to the foundation posts.
- Inside joists shall be 2x6 (min) and shall be connected to the foundation posts using a header board and proper joist hangers.
- There shall be no more than 16" between joists.

- There shall be four total joists: (2) inside and (2) outside.
- All framing must be attached with 3" ring-shanked nails or 3" decking screws.
- The 1st three decking boards at either end of the ramp must be attached with screws.

E. Slopes

- All landings must be level (slope no greater than 1:50).
- All cross slopes must not exceed 1:50.
- Ramps must have a maximum slope of 1:12 with few exceptions.
- In the RARE case a steeper slope is needed, the following will apply:
- A 1:10 slope is permitted for no more than a 6" rise.
- A 1:8 slope is permitted for no more than a 3" rise.

F. Ramps

- Shall not rise more than 30" between landings.
- Shall lead to a firm stable landing, provider is responsible for the area at the end of a ramp.
- The ramp must be finished with concrete a minimum of 1" deep, 12" long, and the width of the ramp to aid in transition to the ground.
- Shall be level with a cross slope of no more than 1:50.
- Shall have a clear width of 41".
- With a drop off of more than 2" shall have a minimum 4" curb.
- Shall allow for water to run off and not collect.
- Must be slip resistant (floor areas painted with a non-slip paint).
- Must use 2'x6" or 5/4"x6" as decking.
- Ramps are not to be built attached to the house unless necessary (2' spacing).
- Ramps for participants using stretchers must avoid turns and be angled if possible.

Note: The end of the ramp must be finished with concrete.

G. Landings (General)

- Must be level – slope less than 1:50, must be stable and slip resistant.
- Landing must have pickets.
- Landings must have top rail, top support rail and curbing.
- Must be at least 60"x60" clear space if the ramp changes direction.
- "Straight through" landings must be the width of the ramp and 5' long.
- "Switchback landings" must be at least 60"x96" clear space.
- Joists must be no more than 16" apart.

H. Landings at Doorways

- Shall have no more than a ½" beveled threshold between landing and interior floor.
- Must have a minimum 18"+ of clear space past the latch side of the door (24" preferred).
- Must minimum have 60"x60" clear space.
- Railings are required for drop offs of more than 2".
- Must not be positioned in such a way as to expose people to the danger of falling, (i.e., railings blocking exposed steps).

I. Handrails

- Must be made from 1½" galvanized pipes (fence top rail) with support blocking every 5'.

- Poles come in 10' lengths and are tapered to fit together.
- Blocking must be screwed into the top rail.
- Shall be attached to blocking with galvanized "C" clamps and glued on PVC caps.
- Shall run parallel to the floor of the landing or ramp.
- Shall be continuous on both sides.
- Shall extend 6" past the end of the last block, but not past the end of the ramp.
- Shall have a 1½" clearance between handrail and top rail and any wall.
- Shall be between 34" and 38" high.
- Shall not rotate within its fittings, drill a screw into galvanized pipes at every block.
- Must be able to withstand 350 lbs. of pressure.

J. Top rails, side rails and curbing

- All lumber above the floor level is to be attached using decking screws.
- Top rail must have a side support rail on the inside of the ramp.
- All rails shall be attached on the inside of the ramp to the 4x4 posts (no toe- nailing between posts).
- 2x2 pickets must be placed no more than 4" apart, must be pre-drilled and installed with a 2½" decking screw, a 2x4 must be substituted for the last picket at the end of the ramp.
- Pickets must be screwed into the top rail, curbing and the ramp frame. Brad nailing pickets into place before screwing can save time.
- Curbing shall be attached on the inside of the ramp to the 4x4 posts (no toe- nailing between posts) and be 2" off the ramp.

K. Heavy Duty Ramps and Landings

- Ramps and landings for participants above 250 lbs. shall be built to accommodate the extra weight.
- These ramps shall use: 2x6 decking boards; 2x8 joists every 16"; and be a minimum of 44" wide clear space.

L. Specs for Hand-Held Showers

- Minimum 6' hose
- Pause/shut off function
- Installation with Teflon tape